Kaustubh Watane

AI Engineer | LLM & RAG Applications

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LinkedIn | GitHub | Portfolio

SUMMARY

AI Engineer at GE HealthCare, building LLM tools for engineers working on regulated medical-device projects. Delivered an on-premise RAG assistant used by more than half of the team, and is now working on a pipeline that generates verification and validation cases ready for compliance review. Interested in retrieval quality, answers that trace back to a source, and AI that works within strict security limits. Also launched an AI companion app that runs on multiple LLM providers, with 500+ users.

EXPERIENCE

GE HealthCare **Aug. 2025 – Present**

AI Engineer (Trainee Engineer) *Bengaluru, India*

Nebula AI – Enterprise AI Workspace *Internal tool*

- Designed and deployed a RAG assistant that lets engineers search thousands of internal specs, procedures, and standards on a regulated network, running fully on-premise with locally hosted models. More than half of the engineers on the team now use it, with a 30%+ gain in workflow efficiency.
- Implemented a four-stage hybrid retrieval pipeline that pairs keyword matching (BM25) with BGE-M3 embeddings in Qdrant, merged with Reciprocal Rank Fusion and re-ranked by a cross-encoder. Keyword search catches part numbers and standard IDs that embeddings blur, and every answer links to its source passage.
- Developed a voice assistant using streaming speech-to-text and text-to-speech over WebSockets. It creates tasks, notes, and calendar events by voice and responds in under a second.
- Expanded the tool from a question-answering assistant into a full workspace with multimodal document understanding, a code assistant, smart notes with autofilled templates, and an admin panel with role-based access and audit logs.
- Initiated integration of Amazon Bedrock, the LLM service approved by GE HealthCare, into both Nebula and TestMatrix.

TestMatrix AI – V&V Test Automation *Internal tool*

- Initiated development of a tool that generates verification and validation test cases for medical-device hardware in the format required by two regulatory standards, IEC 62304 and FDA 21 CFR Part 820, so the team spends its time on review instead of writing from scratch.
- Implemented fast rule-based checks (structure, format, consistency) before and after the LLM step, so bad input never reaches the model and bad output never reaches a reviewer.
- Introduced hallucination detection, a second LLM pass that verifies each generated case, and quality gates, with the aim that reviewers can approve the output instead of repairing it.

SKILLS

Generative AI: LLMs, RAG, Agentic AI, Multi-Agent Systems, LangGraph, LangChain, Prompt Engineering, Tool Calling, Multimodal AI.

Retrieval: Hybrid Search, BM25, BGE-M3 Embeddings, Reciprocal Rank Fusion, Cross-Encoder Reranking, Qdrant, Document Parsing, OCR.

Evaluation: LLM Evaluation, Hallucination Detection, LLM-Based Verification, Guardrails, Quality Gates.

Voice AI: Speech-to-Text, Text-to-Speech, Voice Activity Detection, Real-Time Streaming.

Backend & Data: Python, FastAPI, REST APIs, Async Programming, Pydantic, SQLAlchemy, WebSockets, PostgreSQL, SQL, Redis, Firebase.

Cloud & Deployment: Docker, Ollama, On-Premise AI Deployment, Amazon Bedrock (integration in progress), AWS, Nginx, Linux, Azure AI.

PROJECTS

Luna AI – AI Companion App

Live on Google Play

- Launched a cross-platform AI companion app with chat, voice, journaling, and games, using React Native with Expo and a FastAPI and PostgreSQL backend. It is live on Google Play with 500+ users.
- Designed a thin layer over the Gemini, Claude, Grok, and OpenAI APIs so the app can switch models easily and keep working if one provider goes down.
- Implemented login and sessions and conversation storage in PostgreSQL, and managed releases through Play Store review.

Netra AI – Trading Signal Platform (NSE / NIFTY 50)

github.com/kaustubh812/Netra-AI

- Built a platform that tracks the 50 NIFTY stocks and produces BUY/SELL/HOLD signals by blending an Optuna-tuned XGBoost model with Supertrend, RSI, MACD, and volume signals. Trained on data before 2024 and evaluated on 2024–2025 so the model never sees future information.
- Developed a historical simulation engine that includes real Indian trading costs (brokerage, STT, GST, SEBI charges, stamp duty) and reports Sharpe ratio, max drawdown, and win rate.
- Created the Next.js dashboard with TradingView charts and option-chain PCR tracking, plus an LLM-powered stock screener and a daily market brief.

EDUCATION

C-DAC ACTS, Pune **Aug. 2024 – Mar. 2025**

Post Graduate Diploma in Embedded Systems Design (PG-DESD) *Grade A*

Government College of Engineering, Karad **Jun. 2019 – Jun. 2023**

B.Tech in Electronics and Telecommunication Engineering *CGPA: 7.83/10*

CERTIFICATIONS

- **Building AI Products: Architecture and Orchestration Professional Certificate** – AI system architecture, agent orchestration, and LLM application design.
- **Microsoft Azure AI Essentials Professional Certificate** – Azure AI services, cloud AI/ML deployment, and responsible AI.